little, because the radio observations were not yet detailed enough to guide their theorizing very much. Their only contributions had been the realization that the radio waves are produced by high-speed electrons spiraling around magnetic field lines in the giant radio-emitting lobes, and their calculation of how much magnetic and kinetic energy this entails.

In the 1960s, as the resolutions of radio telescopes continued to improve and optical observations began to reveal new features of the radio sources (for example, the tiny sizes of the light-emitting cores of quasars), this growing body of information became grist for the minds of astrophysicists. From this rich information, the astrophysicists generated dozens of detailed models to explain radio galaxies and quasars, and then one by one their models were disproved by accumulating observational data. This, at last, was how science was supposed to work!

One key piece of information was the radio astronomers' discovery that radio galaxies emit radio waves not only from their giant double lobes, one on each side of the central galaxy, but also from the core of the central galaxy itself. In 1971, this suggested to Martin Rees, a recent student of Dennis Sciama's in Cambridge, a radically new idea about the powering of the double lobes. Perhaps a single engine in the galaxy's core was responsible for *all* the galaxy's radio waves. Perhaps this engine was directly energizing the core's radio-emitting electrons and magnetic fields, perhaps it was also beaming up power to the giant lobes, to energize their electrons and fields, and perhaps this engine in the cores of radio galaxies was of the same sort (whatever that might be) that powers quasars.

Rees initially suspected that the beams that carry power from the core to the lobes were made of ultra-Iow-frequency electromagnetic waves. However, theoretical calculations soon made it clear that such electromagnetic beams cannot penetrate through the galaxy's interstellar gas, no matter how hard they try.

As is often the case, Rees's not quite correct idea stimulated a correct one. Malcolm Longair, Martin Ryle, and Peter Scheuer in Cambridge took the idea and modified it in a simple way: They kept Rees's beams, but made them of hot, magnetized gas rather than electromagnetic waves. Rees quickly agreed that this kind of *gas jet* would do the job, and with his student Roger Blandford he computed the properties that the gas jets should have.